

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka Individual Component

Human Impact and Decision Support System

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Project Proposal Report

Prathibhani Y L O L – IT23317512

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Prathibhani Y L O L	IT23317512	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Floods and landslides remain among the most frequent and damaging natural hazards affecting Sri Lanka, with national records showing over 14.4 million people affected, close to 1.5 million displaced, and 2,218 deaths between 2010 and 2025. Compound flood-landslide events in the central and southern hill country have caused significant human and infrastructure impacts. Existing warning infrastructure in Sri Lanka, including the Disaster Management Centre's Risk Index and situation-monitoring dashboards, provides hazard information but has limited capability to estimate human impact and convert hazard information into prioritized, actionable decisions. This creates a gap between hazard prediction and effective disaster response.

This research proposes a **Human Impact Prediction and Decision Support Component** as part of the **Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka**. The overall system integrates flood and landslide prediction, severity assessment, human impact estimation, and response decision support. The proposed component focuses on the decision layer, receiving hazard probability and severity outputs from the flood and landslide prediction components and combining them with population, demographic vulnerability, and infrastructure accessibility data at the Grama Niladhari division level.

The component computes a composite risk score using a Hazard-Exposure-Sensitivity-Capacity framework, classifies risk into three graded categories, and generates explainable, colour-coded alerts with recommended actions for affected divisions. GeoPandas-based spatial data fusion, Logistic Regression, Random Forest, and XGBoost models will be used, with SHAP integrated to explain prediction results. The methodology follows a quantitative design-science approach with agile iterative development, stratified cross-validation, and historical backtesting using major Sri Lankan disaster events. The resulting outputs will be exposed through an application programming interface and passed to the system's response and resource coordination component to support prioritised emergency actions. Thus, the proposed component contributes to Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka by transforming hazard predictions into locally relevant human-impact assessments and actionable decisions, supporting faster and more effective disaster risk reduction in Sri Lanka.

Keywords: human impact prediction; multi-hazard early warning; explainable machine learning; disaster decision support; geospatial vulnerability assessment

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the Curve
DEM	Digital Elevation Model
DRR	Disaster Risk Reduction
DS Division	Divisional Secretariat Division
DSR	Design Science Research
EOC	Emergency Operations Centre
F1	F1 Score (harmonic mean of precision and recall)
GIS	Geographic Information System
GN Division	Grama Niladhari Division
HDX	Humanitarian Data Exchange
HTTPS	Hypertext Transfer Protocol Secure
IDMERS	Integrated Disaster Management and Early Warning System
JSON	JavaScript Object Notation
MHEWS	Multi-Hazard Early Warning System
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SUS	System Usability Scale
TLS	Transport Layer Security

Table 01. List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Sri Lanka is highly exposed to multiple natural hazards because of its tropical climate, monsoonal rainfall patterns, mountainous terrain, and extensive river systems. Among these hazards, floods and landslides are among the most frequently recurring and damaging events, particularly during periods of intense rainfall [1]. The central highlands, including Ratnapura, Kegalle, and Nuwara Eliya, are repeatedly affected by landslides, while the Kalu Ganga and Kelani Ganga river basins experience seasonal and flash flooding [2]. These hazards can also overlap during intense rainfall, creating complex consequences for communities, infrastructure, and emergency response.

National disaster records from 2010 to 2025 demonstrate the scale of this problem. During this period, recorded disasters affected approximately 14.4 million people, displaced nearly 1.5 million individuals, and resulted in 2,218 deaths across 59 events [2]. Flood events occurring together with landslides accounted for more than 6.4 million affected people, approximately 894,179 displaced individuals, and 479 deaths, making compound flood-landslide events particularly significant from a human-impact perspective [2]. These figures demonstrate that hazard severity alone is insufficient. A hazard affecting densely populated and vulnerable communities with limited infrastructure access can produce greater consequences than a similar hazard in a sparsely populated area.

The impacts of floods and landslides are not distributed equally among communities. Hazard-prone populations differ in population density, age structure, household conditions, economic vulnerability, disability, and access to essential services. Children, older adults, and persons with disabilities may face greater evacuation difficulties, while economically vulnerable households may have fewer options for relocation and recovery [3]. Infrastructure accessibility is another important factor influencing disaster consequences. Roads, bridges, hospitals, schools, emergency shelters, and other essential facilities can become inaccessible or damaged during flood and landslide events. Consequently, two Grama Niladhari divisions experiencing similar hazard severity may require different levels of emergency attention depending on their population characteristics, vulnerability, and access to critical infrastructure. The 2016 Aranayake landslide and major flooding events in Ratnapura and Kegalle demonstrate how consequences vary with location and community characteristics [2].

Sri Lanka's disaster management institutions provide important hazard and risk information through the Disaster Management Centre and other technical agencies. The Disaster Management Centre Risk Index incorporates hazard, population, and infrastructure-related indicators to represent disaster risk across the country [4]. However, this information is primarily intended for broader risk assessment rather than continuously converting multi-hazard severity into location-specific human-impact priorities. Existing early-warning approaches also tend to focus on individual hazards. Flood-related monitoring can provide information such as rainfall, river levels, and flood conditions, while landslide warnings provide information regarding landslide-prone locations and environmental conditions [1].

However, these outputs are not necessarily combined with demographic vulnerability and infrastructure-accessibility information to determine likely human consequences.

Another limitation is the separation between hazard information and response decisions. A warning indicating high flood or landslide severity does not directly indicate how many people may be affected, which population groups may require additional assistance, which locations should receive priority, or what action should be taken first. Research evaluating multi-hazard early-warning systems in South and Southeast Asia has similarly identified weaknesses in risk knowledge and impact-based communication despite comparatively stronger capabilities in hazard detection, monitoring, and forecasting [5]. Static indicators and predetermined classifications are useful for long-term planning but may not represent changing consequences during an actual event. Population exposure, vulnerability, accessibility to infrastructure, and hazard severity need to be considered together to support more meaningful event-specific decisions.

The proposed research addresses this gap through a Human Impact Prediction and Decision Support Component within the Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka. The broader system coordinates hazard prediction, impact assessment, and emergency response. The first component focuses on flood prediction and severity assessment across the Kalu Ganga and Kelani Ganga river basins. The second component focuses on landslide occurrence prediction and forecasting within selected highland areas, including Kothmale, Ambagamuwa, and Walapane. The proposed component receives upstream hazard outputs and combines them with population, demographic vulnerability, and infrastructure accessibility information to estimate human impact and determine response priorities. A subsequent response and resource coordination component uses these priorities for emergency resource allocation.

The proposed Human Impact Prediction and Decision Support Component therefore functions as the decision layer connecting hazard prediction with emergency response. First, flood and landslide severity outputs are spatially integrated with demographic, population, vulnerability, and infrastructure datasets at Grama Niladhari level using GeoPandas and geographic information system techniques. Second, a composite assessment based on the Hazard-Exposure-Sensitivity-Capacity framework is developed to represent potential human impact. Third, Logistic Regression provides a baseline, while Random Forest and XGBoost are evaluated for relationships between hazard conditions, vulnerability indicators, and observed disaster impacts. SHAP-based explainable artificial intelligence is incorporated to identify the factors contributing to individual predictions. Finally, the component generates graded, colour-coded alerts with recommended actions through a documented application programming interface. The study focuses on Ratnapura, Kegalle, Kothmale, Ambagamuwa, and Walapane administrative areas and the Kalu Ganga and Kelani Ganga river basins. Grama Niladhari-level analysis provides more locally meaningful information than broader district assessments, identifying differences between communities according to exposure, vulnerability, and infrastructure accessibility. The methodology follows a quantitative, design-science approach supported by geospatial data fusion, supervised machine learning, stratified cross-validation, and historical backtesting using major Sri Lankan disaster events. This supports targeted and timely emergency decision-making.

1.2 Literature Review

The literature relevant to this component falls into three groups: studies that model hazard susceptibility and risk using statistical or machine learning methods within the Sri Lankan context; studies that apply multi-hazard, decision-oriented artificial intelligence frameworks internationally; and evaluations of the institutional adequacy of early warning systems in Sri Lanka and comparable South and Southeast Asian countries. Each is reviewed in turn, followed by a review of existing operational systems and products.

1.2.1 Hazard Susceptibility and Risk Modelling in Sri Lanka

Hemasinghe et al. [7] applied a logistic regression model to map landslide susceptibility across Badulla District, using slope, aspect, lithology, land cover, and distance from rivers and roads as causative factors obtained from the Survey Department and the Geological Survey and Mines Bureau of Sri Lanka. The study found that land use carried the highest regression coefficient of the retained factors, at 3.55, more than double that of slope, while distance from rivers and distance from roads returned a coefficient of zero and were removed from the final model entirely. Validation against an independent set of observed landslide locations showed that 76 percent of historical landslides fell within the zones the model classified as highly or extremely highly susceptible, a zone covering only 20.5 percent of the district's total area. The method is methodologically sound and its explicit factor-elimination step is a valuable precedent, but the study is limited to landslide susceptibility as a static, terrain-based probability and does not estimate population exposure, demographic vulnerability, or the resulting human impact of a landslide event, nor does it address flooding or the compound interaction between the two hazards. This finding directly informs the proposed feature engineering process for the present research: candidate features are proposed to be tested for genuine predictive contribution and dropped if that contribution proves negligible, rather than retained purely because they appear plausible.

Weerasinghe et al. [8] conducted a qualitative flood risk assessment for the Western Province of Sri Lanka at the Grama Niladhari division level, the same administrative resolution proposed for this research. The study computed a risk index as the product of a Hazard Index, an Exposure Index, and a Vulnerability Index, with vulnerability disaggregated into social, economic, and physical (housing) sub-indices derived from the 2012 Census of Population and Housing. Critically, the study found that economic vulnerability, measured as the proportion of the population that is unemployed or economically inactive, was a materially stronger predictor of flood risk than age-based social vulnerability, with only 0.9 percent of divisions classified as high social risk compared to 7.1 percent classified as high economic risk.

The study also demonstrated that divisions with comparatively low flood hazard still produced severe human consequences where population and housing vulnerability were high, and validated its hazard classification by comparing predicted high-risk divisions in Gampaha and Colombo against the areas actually affected during the May 2016 floods. This study is the closest existing precedent to the present research in both formula structure and validation method, and its finding that hazard severity alone is an unreliable predictor of harm provides direct empirical justification for the human impact layer proposed in this research. Its principal limitation is that it was conducted for a single province using twelve-year-old census data and did not incorporate machine learning, real-time hazard input, or landslide risk, all of which this research proposes to address.

De Silva et al. [2] conducted a sixteen-year retrospective analysis of natural hazard occurrence and human impact across Sri Lanka using the Emergency Events Database, cross-validated against ReliefWeb and the national Disaster Management Information System. In addition to the national totals already cited, the study documented that drought events affected over 6.5 million people nationally while recording zero deaths and zero displacement in the same historical dataset, a pattern the authors attribute to the systematic under-reporting of slow-onset disasters in humanitarian databases rather than to droughts being genuinely harmless. This finding is methodologically important for the present research because it demonstrates that the absence of recorded impact in a historical hazard database, including DesInventar, cannot be treated as equivalent to the absence of real impact, and this consideration is proposed to be addressed by prioritising well-documented, high-profile validation events over reliance on the full historical record in aggregate.

1.2.2 Machine Learning-Based Multi-Hazard Prediction

Karunawardhana et al. [9] proposed an integrated flood and landslide early detection system for Sri Lanka combining a LoRa-based wireless sensor network for flood monitoring with a U-Net convolutional model for satellite-image-based landslide segmentation, and separate tabular machine learning pipelines for predictive risk scoring. Their landslide susceptibility model, which compared Linear Regression as a baseline against Random Forest and Gradient Boosting, achieved an overall accuracy of 84 percent, an F1 score of 0.78, and a receiver operating characteristic area under the curve between 0.85 and 0.89, with rainfall, elevation, latitude, and longitude identified as the most predictive conditioning factors. Their flood prediction pipeline, combining a Random Forest classifier with a Long Short-Term Memory

network for temporal forecasting, achieved 80 and 85 percent accuracy respectively. Notably, their U-Net landslide detection model achieved a headline accuracy of 98.76 percent but a substantially lower recall of only 63.13 percent, meaning the model correctly classified the great majority of image pixels but still failed to identify a meaningful share of the landslide areas that mattered most. This result is directly relevant to the present research, it demonstrates that a single aggregate accuracy figure can conceal weak performance on the minority class that carries the highest operational cost, which is precisely the concern this research holds regarding its own Critical risk category, and it justifies reporting per-class recall, cross-validation stability, and the train-test performance gap in addition to overall accuracy. The study's precedent of benchmarking a simple linear model against ensemble methods before selecting a final classifier is also directly adopted in the proposed methodology of this research.

Lu et al. [10] proposed a GeoAI-driven, decision-oriented methodology for multi-hazard early warning system development, validated through a subdistrict-level flood and typhoon prototype in Shanghai. The methodology's central argument is that the persistent global problem of "early warning without action" arises because most systems remain forecast-centric, delivering hazard magnitude rather than expected impact, and the paper proposes a four-component architecture consisting of stakeholder-specific artificial intelligence agents, an evidence-based decision support mechanism structured around the questions of When, Where, Who, What, and How, a multi-agent deep reinforcement learning loop for continuous adaptation, and a geographic knowledge graph integrated with large language models for explainable reasoning. The study explicitly cautions that relying on freely generated artificial intelligence explanations for safety-critical decisions risks hallucination and non-traceable reasoning, and argues instead for grounded, mathematically auditable explanation methods. This caution directly supports the selection of SHAP, rather than any generated natural-language explanation, as the proposed explainability method for the present research. The methodology's scope and its reliance on dense, real-time IoT sensor infrastructure and large-scale urban data reflect a different operating context from Sri Lanka's data availability, and this research proposes a comparatively lighter, tabular, ensemble-based approach suited to the demographic and historical hazard data available for the country.

1.2.3 Institutional Evaluation of Early Warning Systems

Aguirre-Ayerbe et al. [5] conducted a baseline evaluation of the availability and adequacy of Multi-Hazard Early Warning Systems across the Maldives, Sri Lanka, Myanmar, and the Philippines, using a 115-question assessment instrument derived from the World Meteorological Organization's Multi-Hazard Early Warning Systems checklist and applied through structured interviews with national disaster management stakeholders. The study found that across all four countries, the element concerning detection, monitoring, and forecasting of hazards was consistently the most developed, while disaster risk knowledge and warning dissemination and communication remained the least developed elements. For Sri Lanka specifically, none of the four internationally defined early warning system elements were assessed as fully developed and operational except the existence of monitoring systems, and the study documented an absence of standardised vulnerability data disaggregated by age and disability, insufficient mechanisms to verify that warnings reach at-risk populations, and warnings that do not account for differentiated vulnerability across sub-populations. This is independent, stakeholder-validated evidence, external to Sri Lanka alone, that the specific gap this research addresses, namely the translation of hazard detection into impact-based, vulnerability-aware decision support, is a genuine and documented regional deficiency rather than an assumption. Its limitation for the purposes of this research is that the assessment is qualitative and perception-based, relying on a small number of institutional respondents per country, and does not itself propose or test a technical solution to the gap it identifies.

Hettiarachchi and Weeresinghe [11] examined the evolution of the Sri Lankan early warning system since the 2004 Indian Ocean tsunami through interviews with the Disaster Management Centre, the National Building Research Organisation, and the Sri Lanka Red Cross. The study found that training and coordination between agencies is the strongest lever available for improving early warning effectiveness, that disability prevalence in Sri Lanka is between 6 and 12.6 percent while protocols for the evacuation of disabled persons remain absent from national disaster management policy, and that private sector engagement in disaster risk reduction remains limited. The explicit, documented absence of disability-specific evacuation protocols in Sri Lankan policy is direct justification for including disability rate as a demographic vulnerability factor in the proposed human impact formula, rather than including it as a generic addition without local grounding.

1.2.4 Existing Systems and Products

Three operational systems currently attempt to address parts of this problem in Sri Lanka. The DMC Risk Index combines hazard, population, and infrastructure indicators into a national risk classification and is technically the closest existing precedent to the formula structure proposed in this research, but it is compiled and published as a static report rather than an automated system that responds to a live hazard severity input [4]. A forecast-based warning platform operated jointly by the Disaster Management Centre and the International Water Management Institute provides active, real-time forecasting, but is structured around a single hazard type at a time and does not fuse flood and landslide risk into one shared, population-aware decision layer. A situation-report visualisation dashboard used for national disaster incident monitoring provides live data display for decision-makers, but published evaluation of comparable dashboard-based decision support in the Sri Lankan disaster management context has documented data accuracy and delivery-timeliness limitations during active events, reinforcing the finding of Aguirre-Ayerbe et al. [5] that dissemination and impact communication, rather than raw data availability, is the weaker link in the chain. None of these three systems combines multi-hazard severity, population and vulnerability data, and infrastructure accessibility into a single, explainable, prioritized decision output, which is the specific limitation the present component proposes to address.

1.3 Research Gap

Three converging lines of evidence establish this gap. First, national hazard records demonstrate that human impact varies independently of hazard frequency, with compound cyclone events producing the highest mortality despite comparatively low frequency, and flood-landslide combinations producing the greatest overall affected population [2]. Second, an independent, stakeholder-validated regional evaluation confirms that Sri Lanka's early warning capability is comparatively strong in hazard detection and monitoring but weak in disaster risk knowledge and impact-based communication, with no standardized, disaggregated vulnerability data currently incorporated into national warning processes [5]. Third, existing Sri Lankan systems, including the DMC Risk Index, single-hazard forecast platforms, and situation-report dashboards, each address only a partial slice of this problem, and none fuses multi-hazard severity with demographic vulnerability and infrastructure accessibility into one explainable, prioritized, actionable decision [4], [5].

The specific research gap addressed by this study is stated as follows: there is currently no validated method, grounded in Sri Lanka's own demographic and historical disaster data, for converting multi-hazard flood and landslide severity into an explainable, vulnerability-aware estimate of human impact at the Grama Niladhari division level, and no operational Sri Lankan system converts such an estimate into a graded, actionable early warning decision.

2 OBJECTIVE

2.1 Main Objective

To design, develop, and validate a machine learning-based human impact prediction and decision support component that converts flood and landslide hazard severity into an explainable, vulnerability-aware estimate of human impact and a graded, actionable early warning decision, at the Grama Niladhari division level, for the Ratnapura, Kegalle, Kothmale, Ambagamuwa, and Walapane areas and the Kalu and Kelani river basins, achieving a validated classification performance that improves measurably on a logistic regression baseline when back tested against documented historical disaster events.

2.2 Specific Objectives

1. To design and populate a spatially joined dataset that fuses historical hazard event records from DesInventar, demographic and vulnerability indicators from the 2024 Census of Population and Housing, and infrastructure accessibility data derived from OpenStreetMap, at the Grama Niladhari division level, for the defined coverage area.
2. To formulate and empirically validate a composite human impact scoring formula, structured as **Hazard** × **Exposure** × **Sensitivity** ÷ **Capacity**, with feature weights informed by published Sri Lankan flood and landslide risk literature, and to test each candidate feature for genuine predictive contribution rather than including it by assumption.
3. To design, train, and comparatively evaluate at least three classification approaches, namely a logistic regression baseline, a Random Forest classifier, and an XGBoost classifier, selecting the model that achieves the strongest validated performance on the four-class human impact risk classification (Low, Moderate, High, Critical).
4. To integrate SHAP-based explainability into the prediction pipeline such that every generated alert is accompanied by a traceable, feature-level rationale for its classification

5. To validate the trained model against three documented historical events, namely the 2016 Aranayake landslide, the 2017 monsoon floods, and the 2021 Ratnapura and Kegalle floods, held out entirely from model training, and to report per-class precision, recall, F1 score, and cross-validation stability in addition to overall accuracy.
6. To expose the final decision output, comprising a priority score, a graded warning level, a recommended action, and its SHAP-based explanation, through a documented application programming interface suitable for consumption by the Emergency Response and Resource Coordination component of the IDMERS system.

3 METHODOLOGY

3.1 Research Methodology / Design

This research proposes to adopt a Design Science Research (DSR) methodology, combined with a quantitative, applied machine learning research design. This choice is justified on three grounds. First, the nature of the research question is not purely explanatory but constructive: the objective is to design and validate an artefact, namely a predictive decision support component, rather than only to test a hypothesis about an existing phenomenon, which is the defining characteristic of Design Science Research as applied in information systems and software engineering capstone research. Second, the type of data involved, comprising structured demographic records, historical disaster logs, and hazard severity scores, is inherently quantitative and lends itself to supervised statistical learning rather than qualitative interpretive analysis. Third, the experimental requirement of the study, namely training a classifier and benchmarking it against a baseline using measurable performance metrics, is consistent with the quantitative predictive modelling approach adopted in directly comparable Sri Lankan literature, including Hemasinghe et al. [7] and Karunawardhana et al. [9], both of which apply supervised statistical or machine learning models to Sri Lankan hazard data and validate them against real, historical outcomes. Accordingly, this research proposes to combine the Design Science Research paradigm, for the construction and iterative refinement of the decision support artefact, with a quantitative predictive modelling methodology, for the development and evaluation of the underlying classifier.

3.2 Data / Participants / Experimental Inputs

The research requires four categories of data, each selected because it directly instantiates one term of the proposed Hazard-Exposure-Sensitivity-Capacity formula.

Hazard severity data is received, not independently collected, as the output of the flood and landslide severity models developed by the other members of the IDMERS project team, covering the Kalu and Kelani river basins for flood severity and the Kothmale, Ambagamuwa, and Walapane areas for landslide severity. This data is appropriate because it constitutes the triggering input without which no human impact estimate can be computed.

Demographic and vulnerability data is proposed to be drawn from the Department of Census and Statistics' 2024 Census of Population and Housing, providing population density, age distribution, household counts, and disability rate at the smallest available administrative resolution. Where district-level aggregation is the finest resolution published for a given indicator, such as elderly population percentage, a stated national average is proposed to be used as an explicit, documented assumption pending finer-grained data availability.

Historical hazard impact data is proposed to be drawn from DesInventar Sri Lanka, the disaster information management system maintained by the Disaster Management Centre and the United Nations Development Programme, providing historical records of deaths, displacement, houses destroyed, and people affected, disaggregated by district and year since 1974. This dataset constitutes the ground truth against which the proposed model is trained and evaluated, and named, well-documented events within it, specifically the 2016 Aranayake landslide, the 2017 monsoon floods, and the 2021 Ratnapura and Kegalle floods, are proposed to be treated as held-out validation cases, since De Silva et al. [2] establish that this dataset undercounts slow-onset and less-publicized events.

Infrastructure and accessibility data is proposed to be sourced from OpenStreetMap, providing hospital locations, road network status, and shelter locations, and **administrative boundary data** is proposed to be sourced from the Survey Department of Sri Lanka and the Humanitarian Data Exchange, providing the Grama Niladhari division boundaries used to spatially join all other datasets. Supporting **landslide rainfall threshold data** published by the National Building Research Organization is proposed to be used to cross-validate the plausibility of received landslide severity scores against locally established rainfall-triggering thresholds.

Data preparation is proposed to proceed through a spatial join, implemented using GeoPandas and standard GIS spatial analysis operations, in which every dataset is standardised to the same coordinate reference system and joined to a common Grama Niladhari division boundary layer, producing a single, unified feature table in which each row represents one division and each column represents one feature required by the proposed formula or classifier.

3.3 Baseline / Benchmark

The proposed solution is benchmarked against three points of comparison. The first is a naive hazard-only baseline, in which the classification of risk is derived directly from hazard severity alone, without any population, vulnerability, or infrastructure adjustment, representing the behavior of current single-hazard forecast platforms. The second is a logistic regression classifier trained on the same feature set as the proposed ensemble models, following the precedent established by Hemasinghe et al. [7] and adopted by Karunawardhana et al. [9], both of which use a simpler linear or log-linear model as a baseline before introducing ensemble methods. The third is the existing DMC Risk Index structure itself, used as an institutional benchmark to assess whether the proposed, dynamically computed classification produces materially different, and more current, prioritization than the existing static national index for the same divisions.

3.4 Evaluation Metrics

Metric	Why Relevant	Target / Comparison
Accuracy	Baseline indicator of general classification correctness across four risk classes	Exceed the logistic regression baseline and the naive hazard-only baseline
Per-class precision, recall, F1 (Critical class emphasized)	Overall accuracy can mask poor performance on the rarest, highest-cost class, as demonstrated in Karunawardhana et al. [9]	Recall on the Critical class prioritized over overall accuracy
Backtest match rate against named historical events	Directly measures whether the proposed model would have correctly flagged real, documented disasters	Majority of high-impact historical divisions correctly classified as High or Critical
API response latency	The decision output must be delivered fast enough to be operationally useful during an active event	Response returned within 5 seconds per request
System Usability Scale (SUS) score for the decision-support interface	Measures whether the explanation and alert are usable by a non-technical disaster management official	SUS score of 70 or above

Table 02. Evaluation Metrics

3.5 Validation Strategy

The proposed validation strategy follows accepted practice already demonstrated in the Sri Lankan hazard modelling literature reviewed in Section 1.2. Internal validation is proposed to be conducted through stratified five-fold cross-validation, ensuring that the rare Critical class is proportionally represented in every fold, consistent with the cross-validation approach reported by Karunawardhana et al. [9]. External validation is proposed to be conducted by holding out the 2016 Aranayake landslide, the 2017 monsoon floods, and the 2021 Ratnapura and Kegalle floods entirely from model training and comparing the model's predicted risk classification for each affected division against the actual, documented impact recorded in DesInventar, directly mirroring the validation method used by Weerasinghe et al. [8], who compared their predicted high-risk divisions against the areas actually affected during the May 2016 floods. Where feasible, the resulting alerts and explanations are proposed to be reviewed informally with domain-knowledgeable contacts familiar with Disaster Management Centre or National Building Research Organization reporting practice, to assess face validity of the recommended actions attached to each alert level.

3.6 Development Approach

This research proposes to adopt an Agile, iterative development approach, structured around short development sprints. This approach is justified by three characteristics of the project. First, the nature of the product, a machine learning-based decision support artefact, requires repeated experimentation with feature engineering, model selection, and threshold tuning, which is inherently iterative rather than suited to a single linear specify-build-test cycle. Second, the component depends on hazard severity output from two other team members' components whose interface details may evolve during development, requiring the proposed component to accommodate incremental change to its input contract without a full redesign. Third, the intended users, disaster management officials, are best served by a user-centered, iterative refinement of the alert and explanation interface, allowing early and repeated feedback on interpretability before the interface is finalized. Development is therefore proposed to proceed through a sequence of sprints covering data integration, formula and feature engineering, model development, explainability integration, application programming interface development, and interface refinement, with a working, testable increment produced at the end of each sprint.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of proposed solution and its components

The complete IDMERS solution is proposed to be structured as four cooperating components operating in sequence. The first component is proposed to ingest rainfall, river level, and terrain data for the Kalu and Kelani river basins and output a flood severity score. The second component is proposed to ingest rainfall, slope, and soil data for the Kothmale, Ambagamuwa, and Walapane areas and output a landslide severity score. The individual component addressed in this proposal, the Human Impact Prediction and Decision Support component, is proposed to receive both severity scores and convert them into a graded, explainable, prioritized decision for each affected Grama Niladhari division. The fourth component, Emergency Response and Resource Coordination, is proposed to consume that decision and manage the operational dispatch and routing of response resources. Data is proposed to flow through the system in one direction, from raw environmental observation, through hazard severity estimation, through human impact and decision generation, to operational response, with each component exposing a defined interface to the component that depends on it.

4.2 System diagram of components

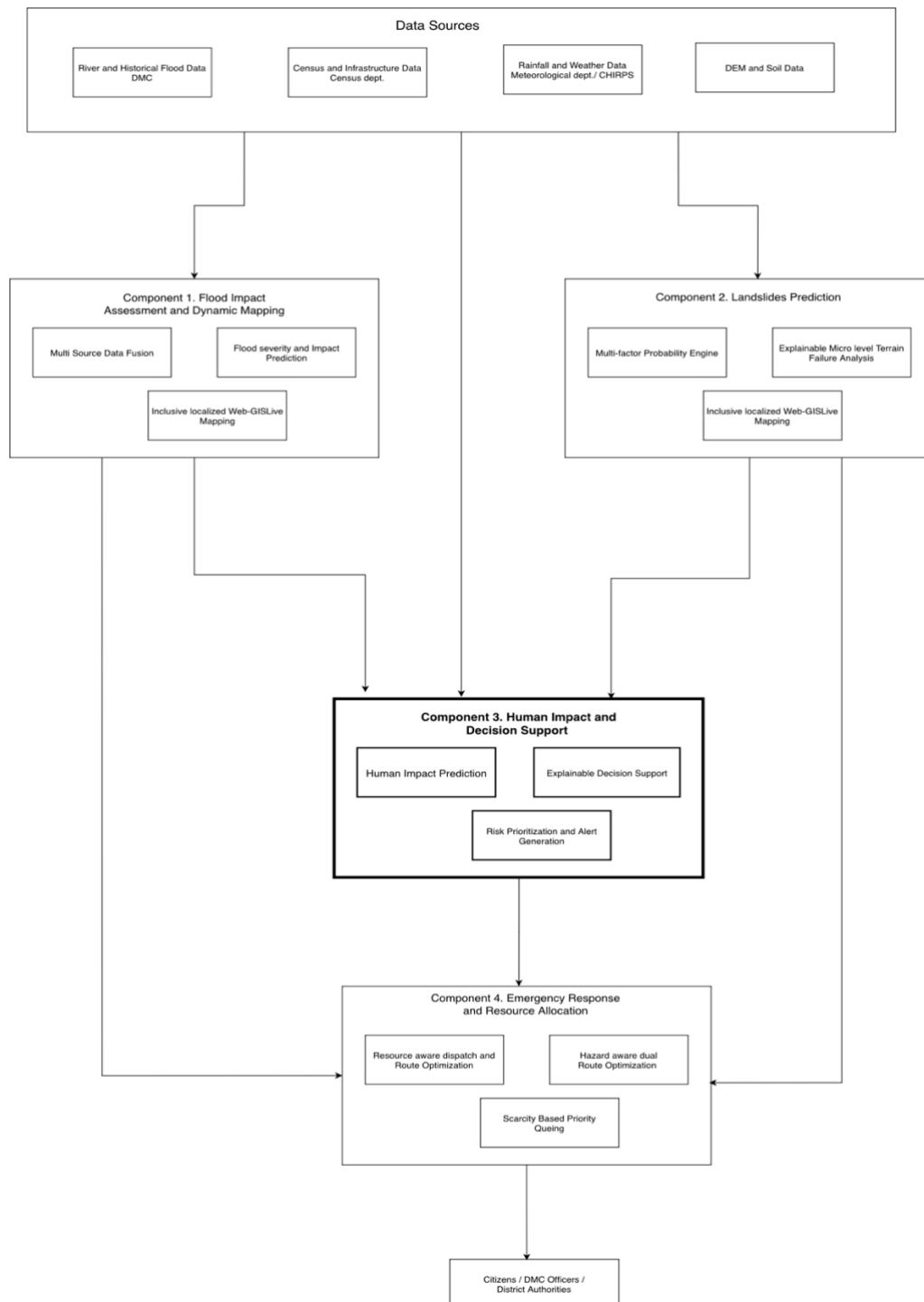


Figure 01. architecture diagram, black-and-white, own component highlighted

4.3 Individual Component

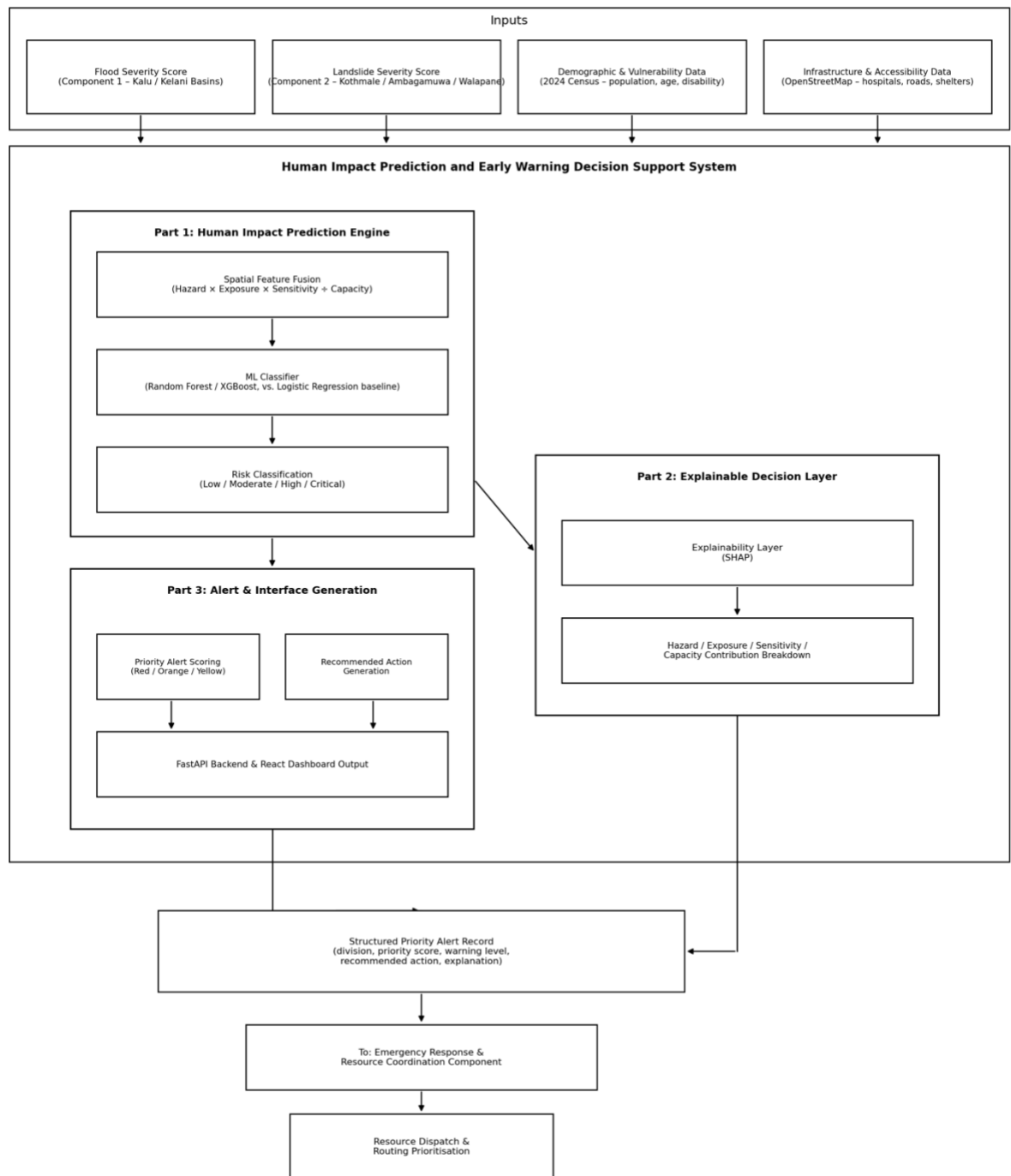


Figure 02. System Architecture Diagram of Human Impact and Decision Support

4.4 Overall Integration

The four IDMERS components are proposed to be integrated through a defined set of RESTful application programming interface contracts rather than shared internal code, so that each component can be developed, tested, and iterated independently. The dependency structure is linear and one-directional for this component: it depends on the flood and landslide severity outputs produced by the first and second components as its required input, and the fourth component depends on this component's output, comprising a priority score, a graded warning level, a recommended action, and an accompanying explanation, as its required input. This component does not recompute severity, and the fourth component does not recompute priority or risk classification; each component owns exactly one stage of the pipeline. The planned integration method is a shared, version-controlled interface specification, defined early in development and treated as a contract between components, with structured JSON as the proposed data exchange format between all four service.

4.5 Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Flood severity modelling (Kalu & Kelani basins)			Sinhalagoda S.A.U.R.A.V
Landslide severity modelling (Kothmale, Ambagamuwa, Walapane)			Bagya R M S
Demographic, vulnerability, and infrastructure data fusion	✓		
Human impact formula design and feature engineering	✓		
Predictive model training, tuning, and validation	✓		
SHAP explainability integration	✓		
Decision and alert-generation logic	✓		
Component API design and development (FastAPI)	✓		
Cross-component interface specification		✓	
System-wide integration testing		✓	
Emergency resource dispatch and routing logic			Palihena M.B
Overall system architecture and final report compilation		✓	

Table 03. Individual Responsibility Boundary

5. USER REQUIREMENTS

5.1 Evidence-Gathering Activities

Stakeholder discussions and domain-expert consultation: A formal data request letter was sent to the Disaster Management Centre (DMC) requesting historical disaster datasets, and a separate request was sent to the Department of Census and Statistics requesting population and vulnerability data relevant to the landslide, decision-support, and emergency-response components. Both institutions responded by directing the request to their existing publicly available online data portals rather than providing a separate dataset, confirming that the required historical disaster records and demographic data are accessible through DesInventar Sri Lanka and the Department of Census and Statistics' published 2024 Census releases respectively, and that no additional institutional data-sharing agreement is required to proceed with the data acquisition described in Section 3.2.

Domain-expert/literature evidence: Weerasinghe et al. [8] found that economic vulnerability, measured as the proportion of the population unemployed or economically inactive, was a substantially stronger predictor of flood risk than age-based social vulnerability, with only 0.9% of Grama Niladhari divisions classified as high social risk compared to 7.1% classified as high economic risk. This directly informs the requirement that the Sensitivity term of the proposed impact formula incorporate economic vulnerability alongside age and disability indicators, rather than age-based indicators alone. Separately, Karunawardhana et al. [9] reported a landslide detection model achieving 98.76% overall accuracy but only 63.13% recall, meaning the model still missed a meaningful share of the cases that mattered most despite an apparently strong headline figure. This directly informs the non-functional requirement below that per-class recall, not overall accuracy alone, be reported and monitored, particularly for the Critical risk category.

Design-thinking / accessibility consideration: Reviewing the Disaster Management Centre's existing Risk Index and situation-report dashboard practice against the findings of Aguirre-Ayerbe et al. [5], who documented that Sri Lankan early warnings do not adequately account for the differing risks and needs of subpopulations, including women, older people, youth, and people with disabilities, confirmed that no existing Sri Lankan system currently tailors its alert content to these groups. This motivates the vulnerability-weighted alert and localisation requirements below.

5.2 Translated System Requirements

Functional Requirements

1. The system shall ingest flood severity and landslide severity scores from the upstream hazard models via a defined API contract, without recomputing either score.
2. The system shall spatially join incoming severity scores with population density, age distribution, disability rate, and household count data at the Grama Niladhari division level.
3. The system shall compute a composite human impact score for each affected division using a formula incorporating both demographic and economic vulnerability indicators.
4. The system shall classify predicted human impact into discrete risk categories (Low, Moderate, High, Critical), consistent with established risk-classification conventions.
5. The system shall generate an explainable, SHAP-based breakdown of each prediction, showing the relative contribution of hazard severity, exposure, sensitivity, and capacity factors to the resulting classification.
6. The system shall display predictions at Grama Niladhari division granularity for the pilot area (Ratnapura, Kegalle, Kothmale, Ambagamuwa, Walapane).
7. The system shall support alert content in Sinhala, Tamil, and English.
8. The system shall output a structured, geotagged decision record (division, priority score, warning level, recommended action, explanation) consumable by the Emergency Response and Resource Coordination component via a defined API contract.
9. The system shall allow authorized users (e.g., DMC/NBRO officers) to review historical prediction accuracy against confirmed disaster impact records for a given division.

Non-Functional Requirements

1. **Response time:** The system shall return a priority classification and explanation for a queried division within 5 seconds under normal network conditions.
2. **Reliability:** The prediction model shall report per-class recall and F1 score, not overall accuracy alone, for each risk category, consistent with the concern raised by Karunawardhana et al. [9], where a 98.76% accuracy model still achieved only 63.13% recall.
3. **Validation:** The model shall be evaluated through stratified cross-validation and backtested against spatially and temporally held-out historical events (the 2016 Aranayake landslide and the 2021 Ratnapura and Kegalle floods) rather than training-region performance alone.
4. **Usability:** At least 90% of test users unfamiliar with the underlying model shall be able to identify a division's current risk level and recommended action within 2 minutes, without prior training.
5. **Accessibility:** The alert interface shall present content in a form usable by officials and residents with differing vulnerability levels, addressing the gap documented by Aguirre-Ayerbe et al. [5] regarding warnings that do not account for subpopulation-specific needs.
6. **Scalability:** The system shall support prediction requests across all Grama Niladhari divisions in the pilot coverage area without response time exceeding the 5-second threshold specified above.
7. **Security:** All data exchanged with the flood, landslide, and resource-coordination components via the API shall use authenticated, encrypted (HTTPS/TLS) connections.
8. **Interoperability:** The decision-output schema shall remain stable and versioned, such that changes to this component's internal model do not break the Emergency Response component's ingestion pipeline.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target market for this component consists of government and quasi-government disaster management entities in Sri Lanka, particularly the Disaster Management Centre at both the national and district levels, the National Building Research Organization, and the Divisional Secretariats in the Ratnapura, Kegalle, Kothmale, Ambagamuwa, and Walapane regions. **The secondary target market** includes international and local humanitarian NGOs involved in disaster response in Sri Lanka, specifically those already engaged in localized risk assessment efforts, who need prioritized and understandable risk data to effectively plan resource pre-positioning and distribution of relief.

The primary customer persona is a **District Disaster Management Officer**, responsible for coordinating response actions across multiple Grama Niladhari divisions during an active flood or landslide event, who currently lacks a fast, consolidated view of which divisions require the most urgent response and why, and who needs that information delivered in a form that can be understood and acted on within minutes, under significant time pressure.

6.2 Value Proposition

The proposed component converts raw hazard severity, which existing single-hazard forecast platforms already provide, into a locally grounded, explainable estimate of human impact and a prioritized, actionable decision, removing the need for an official to manually cross-reference hazard alerts against separate population, vulnerability, and infrastructure records during an active event. Unlike the existing DMC Risk Index, the proposed output updates automatically in response to live severity input rather than requiring periodic manual compilation, and unlike existing single-hazard forecast platforms, it is proposed to address flood and landslide risk within the same decision pipeline for the overlapping Ratnapura and Kegalle areas.

6.3 Revenue Model

A business-to-government licensing model is proposed as the primary revenue pathway, structured as an annual service subscription per participating district or administrative division, covering hosting, maintenance, and periodic model retraining. A complementary grant-funded deployment pathway is proposed for initial rollout, given that comparable Sri Lankan disaster risk reduction initiatives, including area-based risk assessment work conducted by international humanitarian organizations in the country, have historically been funded through donor and development partner grants rather than direct government procurement in their pilot phase. A tiered application programming interface access model is proposed for non-governmental organization users, distinguishing between free, rate-limited access for research and evaluation purposes and a paid tier for organizations requiring guaranteed uptime and higher request volume.

6.4 Cost Estimation

Recurring costs are proposed to comprise cloud hosting for the API and database layer, periodic model retraining as new demographic or hazard data becomes available, and ongoing maintenance of the data integration pipeline as source data formats change. These costs are detailed with specific estimates in the Budget and Justification section of this proposal.

6.5 Pricing Strategy

A freemium pilot pricing strategy is proposed, in which initial deployment to the Disaster Management Centre and district authorities within the defined coverage area is offered at no direct cost during a pilot evaluation period, transitioning to a paid annual subscription per district following successful pilot validation, consistent with typical adoption pathways for government-facing disaster management software in similar contexts.

6.6 Competitive Advantage

The proposed component's competitive advantage rests on three characteristics not jointly present in any existing Sri Lankan system reviewed in Section 1.2: it is the only reviewed approach proposed to fuse multi-hazard severity with demographic vulnerability and infrastructure accessibility into a single decision output; it is the only reviewed approach proposed to generate a traceable, SHAP-based explanation for each classification rather than an opaque score; and unlike the existing DMC Risk Index, it is proposed to operate as a live, automatically updating service rather than a static, periodically published report.

6.7 Intellectual Property Considerations

The trained model weights, the specific feature engineering pipeline, and the calibrated formula parameters developed through this research are proposed to be treated as the intellectual property of the research team, while all underlying public data sources, including the Census of Population and Housing, DesInventar, and OpenStreetMap, are proposed to be used and attributed strictly in accordance with their respective public data licensing terms. A dual-licensing approach is proposed as a long-term intellectual property strategy, in which the core data integration and formula methodology is released under an open academic license to support reproducibility and further research, while a hosted, maintained, and supported deployment of the service remains a proprietary offering to institutional customers.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Development workstation (existing, team-owned)	0	Model training, data processing, and interface development performed on existing personal devices; no new hardware purchase required for the academic phase.
Mobile/tablet test device (existing personal device)	0	Needed to validate the React-based dashboard's usability on smaller screens, consistent with the accessibility requirement in Section 5.

Table 04. Hardware Costs

7.2 Software Licenses

Item	Estimated Cost (LKR)	Justification
Python ML libraries (scikit-learn, XGBoost, SHAP)	0	All core modelling and explainability libraries are open source with no licensing fee.
GeoPandas / GIS spatial analysis libraries	0	Used for spatial join of severity scores with Census, DesInventar, and OpenStreetMap data; no license cost.
FastAPI / React	0	Backend and frontend frameworks are open source; no license cost.
PostgreSQL + PostGIS	0	Version control and collaboration, covered under GitHub's free/student tier.

Table 05. Software Costs

7.3 Cloud Services

Item	Estimated Cost (LKR)	Justification
Cloud VM hosting for the API and PostGIS database (small instance)	3,000–6,000 / month during active development and integration testing	Required so the flood and landslide components can query this component's live decision-generation API during team integration testing, rather than relying on local-only testing that cannot be accessed by teammates
Object storage for prediction logs, SHAP explanation records, and trained model versions	1,000–2,000 / month	Required to satisfy FR9 (Section 5.2), which requires every prediction, its input features, and its explanation to be logged for later audit and model retraining; also needed to persist successive trained model versions (baseline, Random Forest, XGBoost) for reproducibility.
Domain name and SSL/TLS certificate for the API endpoint	5,000 (one-time, annual renewal thereafter)	Required to satisfy the non-functional security requirement (Section 5.3) that all data exchanged with other components use authenticated, encrypted (HTTPS/TLS) connections.

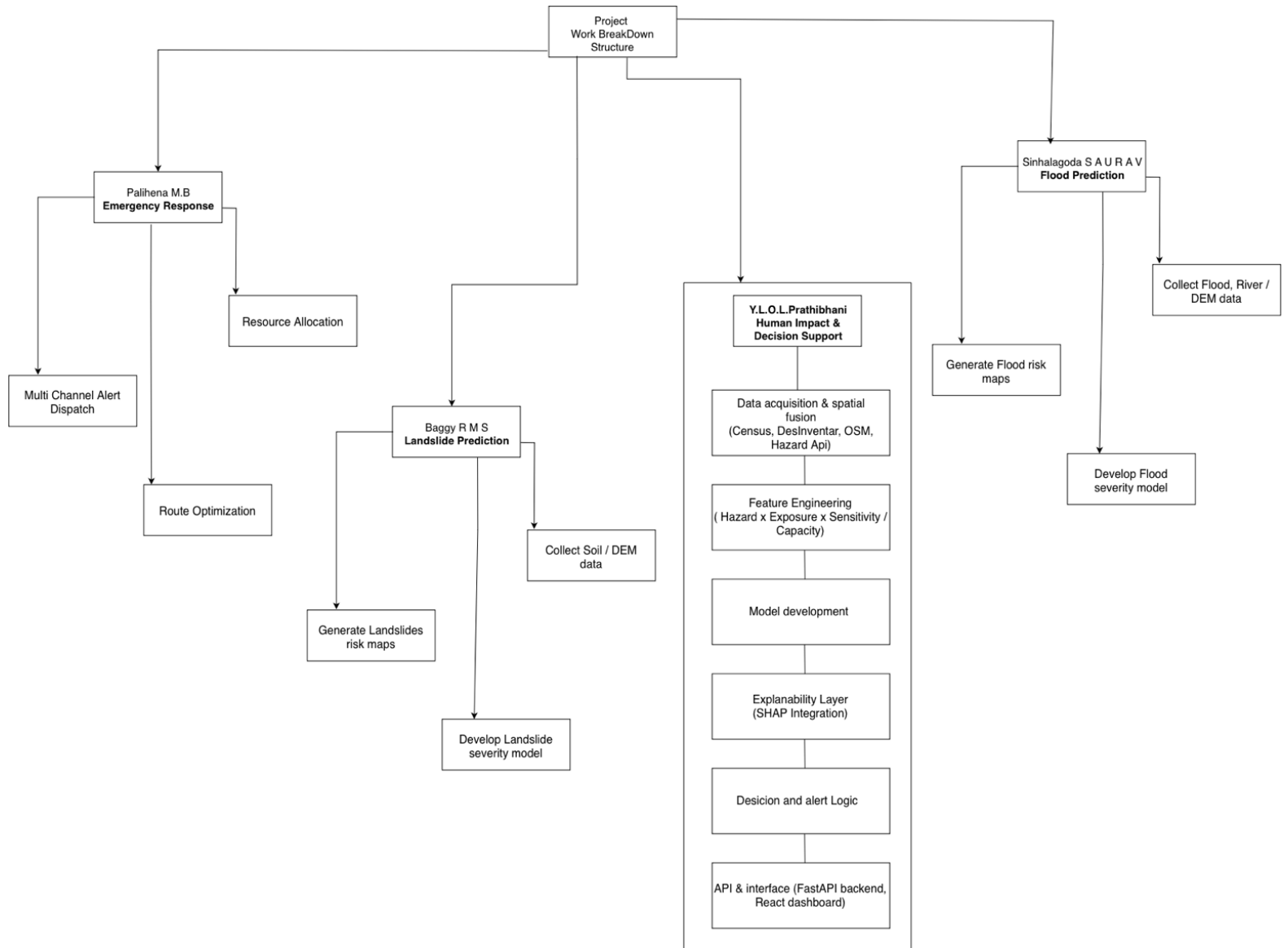
Table 06. Cloud Services

7.4 Summary Justification

The budget is deliberately weighted toward cloud hosting rather than hardware or licensing, since this component's entire technology stack, from geospatial data fusion through model training to API delivery, relies on open-source tooling, consistent with keeping the solution replicable and low-cost for eventual government adoption (Section 6.4, Cost Estimation). GPU compute is budgeted only as a contingency rather than a default cost, since Random Forest and XGBoost on structured, tabular features do not require GPU acceleration, unlike deep-learning approaches used elsewhere in the IDMERS project. The two recurring real expenses are cloud hosting, necessary because the Emergency Response and Resource Coordination component depends on querying this component's decision-generation API live during integration testing rather than through static exports, and object storage, necessary to satisfy this component's own audit-logging functional requirement (FR9) rather than being an optional addition.

8. WORK BREAKDOWN STRUCTURE (WBS)

Figure 03. Work Break Down Chart For Whole System



9. GANTT CHART

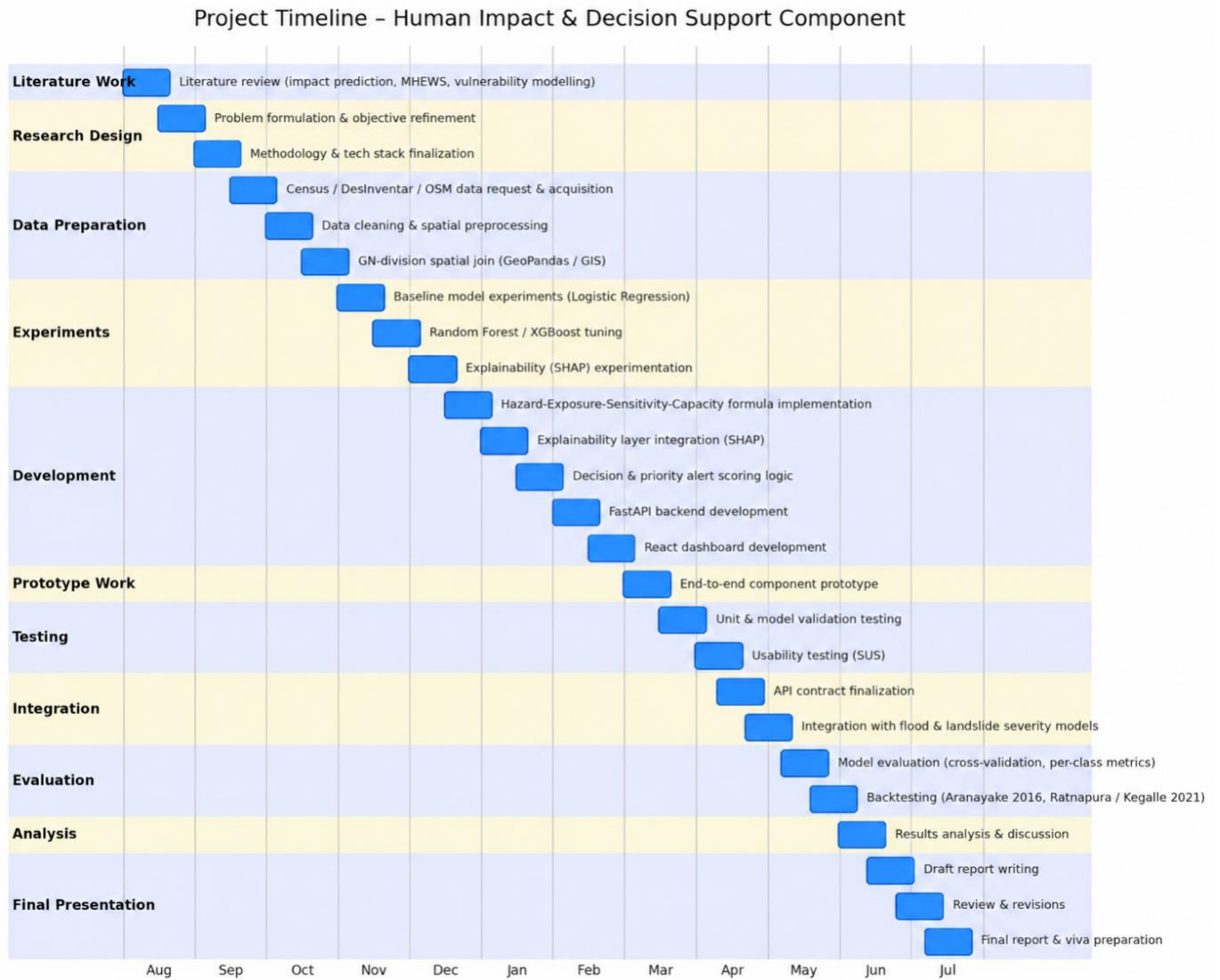


Figure 04. Ganaat Chart

10.REFERENCES

- [1] R. Basher, “Global early warning systems for natural hazards: systematic and people-centred,” *Philos. Trans. R. Soc. Math. Phys. Eng. Sci.*, vol. 364, no. 1845, pp. 2167–2182, Aug. 2006, doi: 10.1098/rsta.2006.1819.
- [2] D. K. M. De Silva, S. S. P. Warnakulasuriya, and S. D. Dharmaratne, “Natural hazards and population impact in Sri Lanka, 2010–2025,” *BMC Public Health*, vol. 26, Art. no. 2335, 2026.
- [3] S. S. L. Hettiarachchi and S. Weeresinghe, “Achieving disaster resilience through the Sri Lankan early warning system: Good practices of disaster risk reduction and management,” *Procedia Econ. Finance*, vol. 18, pp. 789–794, 2014.
- [4] Disaster Management Centre, Ministry of Defence, Sri Lanka, *Disaster Risk Index for Sri Lanka*, 1st ed. [Online]. Available: <https://www.dmc.gov.lk>
- [5] I. Aguirre-Ayerbe, M. Merino, S. L. Aye, R. Dissanayake, F. Shadiya, and C. M. Lopez, “An evaluation of availability and adequacy of multi-hazard early warning systems in Asian countries: A baseline study,” *Int. J. Disaster Risk Reduct.*, vol. 49, Art. no. 101749, 2020, doi: 10.1016/j.ijdrr.2020.101749.
- [6] United Nations Office for Disaster Risk Reduction (UNDRR), *Sendai Framework for Disaster Risk Reduction 2015–2030*. Geneva, Switzerland, 2015.
- [7] H. Hemasinghe, R. S. S. Rangali, N. L. Deshapriya, and L. Samarakoon, “Landslide susceptibility mapping using logistic regression model (a case study in Badulla District, Sri Lanka),” *Procedia Eng.*, vol. 212, pp. 1046–1053, 2018, doi: 10.1016/j.proeng.2018.01.135.

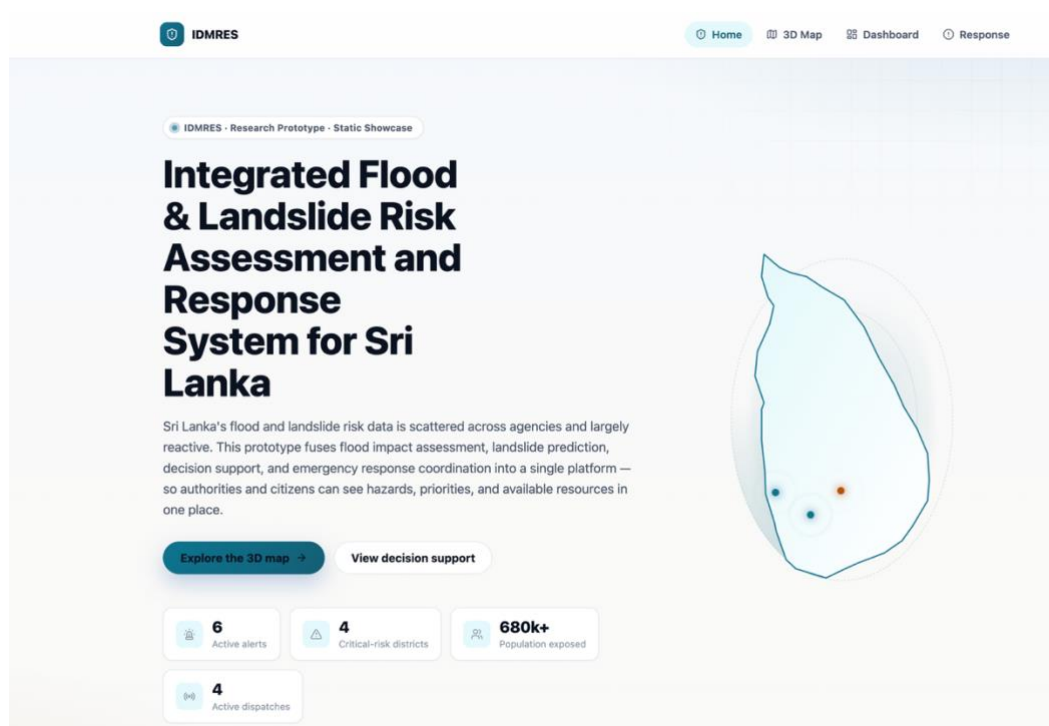
- [8] K. M. Weerasinghe, H. Gehrels, N. M. S. I. Arambepola, H. P. Vajja, J. M. K. Heratha, and K. B. Atapattu, “Qualitative flood risk assessment for the Western Province of Sri Lanka,” *Procedia Eng.*, vol. 212, pp. 503–510, 2018, doi: 10.1016/j.proeng.2018.01.065.
- [9] K. P. I. Karunawardhana, E. M. H. K. B. Ekanayake, S. P. Munasinghe, N. C. Amarasena, and J. G. S. Samiru, “Enhancing human safety through early detection of floods and landslides in Sri Lanka,” in *Proc. 2024 9th Int. Conf. Inf. Technol. Res. (ICITR)*, 2024, doi: 10.1109/ICITR64794.2024.10857761.
- [10] Q. Lu, J. Wen, J. Yan, Y. Wang, and H. Chen, “A GeoAI-driven and decision-oriented methodology for multi-hazard early warning system development,” *Int. J. Disaster Risk Sci.*, 2026.
- [11] Department of Census and Statistics, Sri Lanka, *Census of Population and Housing 2024*. Colombo, Sri Lanka, 2024. [Online]. Available: <https://www.statistics.gov.lk>. Accessed: Sep. 15, 2026.
- [12] Disaster Management Centre and United Nations Development Programme, *DesInventar Sri Lanka—Disaster Information Management System*. [Online]. Available: <https://www.desinventar.net>
- [13] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774.
- [14] Disaster Management Centre and Department of Census and Statistics, “Data request correspondence,” GitHub, 2026. [Online]. Available: [GitHub repository](#)
- [15] P. Y. L. O. L., “J26-IT-334: Human Impact & Decision Support Component documentation,” GitHub repository, 2026. [Online]. Available: [GitHub repository](#)

11.APPENDICES

Appendix 1

Appendix A: Prototype Evidence

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka Live demo:
<https://sri-lanka-3-d-map.vercel.app/>



DECISION SUPPORT

Early Warning & Priority Risk Overview

Joint flood and landslide probabilities, population exposure, and infrastructure accessibility combined into a single priority score per district, translated into graded alerts with recommended actions.

Active alerts

6

Critical-risk districts

4

Population exposed

680,000

Avg. model confidence

87%

Joint risk priority by district

Higher score = higher combined flood + landslide priority



Issued alerts

- Colombo** Extreme 121 hours ago

Rising flood risk detected across Colombo.
Issue evacuation order for exposed low-lying/slope zones
- Kalutara** Extreme 124 hours ago

Rising flood risk detected across Kalutara.
Issue evacuation order for exposed low-lying/slope zones
- Ratnapura** Extreme 127 hours ago

Rising flood risk detected across Ratnapura.
Issue evacuation order for exposed low-lying/slope zones

District risk register

Full breakdown of flood and landslide exposure feeding the priority score above.

DISTRICT	FLOOD RISK	LANDSLIDE RISK	POPULATION EXPOSED	PRIORITY SCORE	RECOMMENDED ACTION
Colombo	Extreme	—	232,000	87	Issue evacuation order for exposed low-lying/slope zones
Kalutara	Extreme	—	121,000	87	Issue evacuation order for exposed low-lying/slope zones
Ratnapura	Extreme	—	69,000	87	Issue evacuation order for exposed low-lying/slope zones
Nuwara Eliya	—	Extreme	47,000	87	Issue evacuation order for exposed low-lying/slope zones
Gampaha	High	—	168,000	62	Issue amber advisory; monitor rainfall thresholds closely
Kegalle	Medium	—	43,000	36	Routine monitoring; no active alert required

Appendix B: Preliminary Research Notes (Requirements Elicitation & Research Design)

Notes	Date	No
Datasets to get		
* Population density		
* Age distribution		
note:- elderly / not available		
* Disability rate.	NBRO rainfall threshold	
Economic Activity	↓	
Hospital location.	cross check lands slide	
Road Network Status	Severity	
Shelter location		
Historical disaster Records		
° Aranyake landslide 2016		
° Rothnapura Kegalle Flood. 2021		
° 2017 Monsoon Flood		

Alert Logic	Date	No
Risk score → Class → Colour → Action.		
Low → Monitor Only		
Moderate → Public Advisory		
High → Prepare Response Resources		
Critical → Immediate evacuation + dispatch priority		
Alert need		
Priority score	SHAP - Game theory Based	
warning level		
recommended Action.		
Explanation (Why)	* LIME - Approximation	
Phase Plan		
Phase 1		

Logistic Regression	Date	No
g) Simple Baseline, interpretable		
* Breiman - style precedent:-		
Simple models sometimes generalize better		
Human Impact Priority Score		
$Risk = Hazard \times Exposure \times Sensitivity \div Capacity$		
→ Formula structure at GN division level		
$(Hazard Index \times Exposure Index \times Vulnerability Index)$		
Note		
Research Paper :- Wernigle et. al May 2016 Flood		
this tell us Hazard Warning is not enough		
low hazard + high vulnerability sometime severe outcome		

Appendix 2 — Mandatory Proposal Evidence and Decision Log

Each student should maintain the following log throughout proposal development.

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion

Table 07. Appendix 2

Appendix 3 — AI Use Disclosure

Students should disclose any use of generative AI tools during proposal preparation.

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
Claude AI	To brainstorm and check research ideas	Ideas for component design and workflow structure	Output is correct because of some parts that I have taken from International Org and National Org and AI gave me the Resources That They Have Didn't Provide external Sites	
ChatGPT	To fix grammar and improve writing	Corrected sentences and fixed grammatical errors in draft paragraphs	Read through all updated text to ensure the original meaning and technical terms remained correct.	

Table 07. Appendix 3